

avdmanager

The `avdmanager` is a command-line tool that lets you create and manage Android Virtual Devices (AVDs) from the command line. An AVD lets you define the characteristics of an Android handset, Wear OS watch, or Android TV device that you want to simulate in the Android Emulator.

If you're using Android Studio, then you don't need to use this tool and you can instead create and manage AVDs from the IDE ([/studio/run/managing-avds](#)).

The `avdmanager` tool is included in the Android SDK Command-Line Tools package at `android_sdk/cmdline-tools/version/bin/avdmanager`.

Syntax

To use `avdmanager`, use the following syntax:

```
avdmanager [global options] command [command options]
```

Global options

Table 1. List of global options for avdmanager.

Global option	Description
<code>-s</code> <code>--silent</code>	Silent mode: only errors are printed out.
<code>-h</code> <code>--help</code>	Usage help.
<code>-v</code> <code>--verbose</code>	Verbose mode: errors, warnings, and informational messages are printed.
<code>--clear cache</code>	Clear the SDK Manager repository manifest cache.

Commands and command options

Table 2. List of commands and options for avdmanager.

Command and options

create avd **-n** *name* **-k** "sdk_id" **[-c {path|size}] [-f] [-p path]**

Create a new AVD. You must provide a *name* for the AVD and specify the ID of the SDK package use for the AVD using *sdk_id* wrapped in quotes. For example, the following command creates an AVD named **test** using the x86 system image for API level 25:

```
avdmanager create avd -n test -k "system-images;android-25;google_api" [-c path/to/sdcard/ -f] [-p path]
```

The following describes the usages for the other options:

- **-c {path|size}**: The path to the SD card image for this AVD or the size of a new SD card image to create for this AVD in KB or MB, denoted with **K** or **M**. For example, **-c path/to/sdcard/** or **-c 1000M**.
- **-f**: Force creation of the AVD. Use this option if you need to overwrite an existing AVD with a new AVD using the same name.
- **-p path**: Path to the location where the directory for this AVD's files will be created. If you specify a path, the AVD is created in **~/.android/avd/**.

delete avd **-n** *name*

Delete an AVD. You must specify the AVD with *name*.

move avd **-n** *name* **[-p path] [-r new-name]**

Move or rename an AVD. You must specify the AVD with *name*. The following describes the usages for the other options:

- **-p path**: The absolute path to the location at which to create the directory where this AVD's files will be moved. If you don't include this argument, the AVD won't be moved. You might choose not to include this argument if you want to rename the AVD in place.
- **-r new-name**: The new name of the AVD being renamed.

list **[target|device|avd] [-c]**

List all available targets, device definitions, or AVDs. If you don't specify **target**, **device**, or **avd**, **avdmanager** lists all three. Include the **-c** argument to receive a compact output suitable for scripts. The **-c** argument is not available when listing all three options together.

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